

High-Temperature Mechanisms

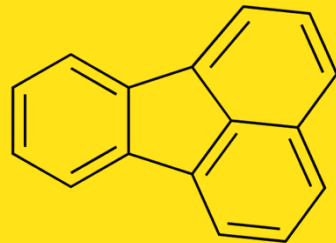
Low-Temperature Mechanism

Phenyl Addition-
DehydroCyclization (PAC)

Hydrogen Abstraction-
Acetylene Addition (HACA)

Hydrogen Abstraction-
Vinylacetylene Addition
(HAVA)

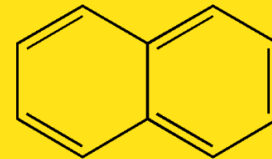
Ryugu PAHs



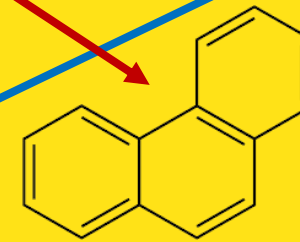
Fluoranthene



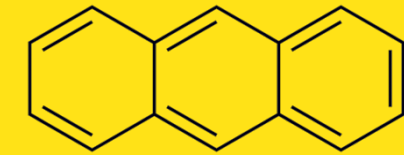
Pyrene



Naphthalene



Phenanthrene

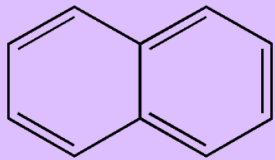


Anthracene

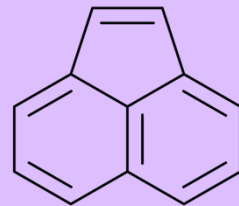
TMC-1 Detected PAHs

Low-Temperature Ryugu Isotopes

High-Temperature Ryugu Isotopes



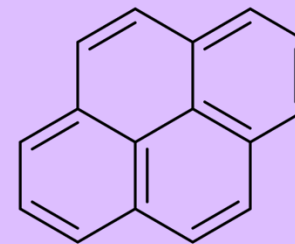
Naphthalene



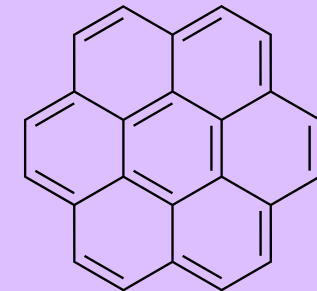
Acenaphthylene



Phenalene



Pyrene



Coronene